PRIME-Factory: An Integrated Predictive Maintenance Framework for Energy-Aware Manufacturing

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Abstract

Modern manufacturing faces a critical challenge: unplanned equipment failures cause costly downtime, waste energy, and increase carbon emissions. While predictive maintenance (PdM) has shown promise in research settings, most solutions remain isolated from energy management, production economics, and auditable decision-making. This paper presents PRIME-Factory, an integrated simulation platform that bridges this gap by unifying AI-driven anomaly detection, health indexing, energy deviation tracking, and factory-level maintenance policy evaluation within a single, transparent framework.

The platform employs Isolation Forest for unsupervised anomaly detection, a Health Index (HI) for degradation tracking, and an Energy Consumption Index (ECI) to capture hidden mechanical inefficiency. Four maintenance policies - Corrective, Preventive, Predictive, and Predictive + Peak Shaving - are benchmarked in a Smart Multi-Product Packaging Factory simulation.

Results demonstrate that predictive policies reduce downtime from 181 minutes to 10 minutes, improve OEE from 50.99% to 95.26% and lower total cost from $1,089 to $343 per event. The addition of Peak Shaving reduces energy consumption by 4.8% with a modest OEE trade-off.

Contextualized for Egypt's strategic fertilizer industry, the framework delivers annual savings of $496,000 per plant, a 5-month payback period, and an NPV of $1.54 million. The platform supports ISO 50001 compliance, workforce upskilling, and Egypt's national climate goals. A tiered deployment model (Tier 1-3) ensures accessibility for SMEs, which constitute over 90% of Egypt's industrial base.

Keywords: Predictive Maintenance; Industry 4.0; Anomaly Detection; Energy Management; Health Index; Peak Shaving; Egyptian Fertilizer Industry; Digital Twin; ISO 50001; SMEs.

1. Introduction

1.1 The Industrial Problem

Modern manufacturing, particularly in energy-intensive sectors, faces a critical paradox. While data generation has exploded due to the proliferation of sensors and IoT, most factories still rely on corrective maintenance (fixing after failure) or rigid preventive maintenance (servicing on a fixed schedule). Corrective maintenance leads to costly unplanned downtime, while preventive maintenance often services healthy assets, wasting spare parts and labor. Meanwhile, energy consumption - often the largest variable cost - is treated as a separate concern rather than a direct indicator of mechanical health. A motor consuming 5% more power under the same load is rarely flagged as a maintenance issue, yet it is often the first sign of bearing wear or friction.

1.2 The Research Gap

The literature provides strong evidence that predictive maintenance can be valuable. Natanael and Sutanto (2022) reported a 13.10% OEE increase and a 62.38% reduction in unplanned failures in a packaging application. Mateo et al. (2025) demonstrated that unsupervised anomaly auditing could improve F1 scores by 192.5% in packaging machinery. However, these studies treat AI, maintenance, and energy as separate silos.

1.3 PRIME-Factory's Contribution

This paper proposes PRIME-Factory, an open-architecture simulation platform designed to bridge this gap. It models a Smart Multi-Product Packaging Factory and demonstrates the full maintenance decision chain: SENSE → CONTEXT → DETECT → CONFIRM → HEALTH → RUL → DECIDE → ACTION → OUTCOME. The platform employs:

- AI (Isolation Forest) for unsupervised anomaly detection.
- Health Index (HI) and Remaining Useful Life (RUL) for degradation tracking.
- Energy Consumption Index (ECI) to detect mechanical inefficiency.
- Four Maintenance Policies (Corrective, Preventive, Predictive, and Predictive + Peak Shaving).

The novelty lies not in a single algorithm, but in the systemic integration of these tools into a decision-support system that provides explainable recommendations (XAI), an economic Cost-Benefit Analysis, and a clear path for industrial deployment.

1.4 Research Questions

#	Question

RQ1	Can unsupervised anomaly detection (Isolation Forest) identify simulated degradation using predominantly healthy training data?
RQ2	Can a Health Indicator (HI) provide more stable maintenance signals than single-point alarms?
RQ3	Does the Energy Condition Indicator (ECI) add useful evidence for detecting degradation-related inefficiency?

2. The Egyptian Industrial Context and the Case for Predictive Maintenance

2.1 The Landscape of Egyptian Industry

Egypt's industrial sector is the backbone of its economy, contributing significantly to GDP, exports, and employment. Key industries include fertilizers, petrochemicals, cement, steel, food processing, and pharmaceuticals. These sectors share a common characteristic: they rely heavily on large rotating equipment—compressors, pumps, conveyors, and motors—that are critical to production continuity.

However, Egyptian industry currently faces several interrelated challenges:

- **Energy Intensity and Cost:** Energy represents a substantial portion of operating costs, particularly in energy-intensive industries. With industrial electricity tariffs now ranging from EGP 1.74 to 2.34 per kWh, and natural gas prices under continuous review, reducing energy consumption is no longer an environmental aspiration but a financial imperative.
- **Unplanned Downtime:** Equipment failures cause production losses that ripple through supply chains. In continuous process industries (e.g., fertilizers, petrochemicals), a single unplanned shutdown can cost millions of EGP in lost production and restart expenses.
- **Lack of Predictive Tools:** Most Egyptian factories still rely on corrective or rigid preventive maintenance. They lack the tools to detect early degradation indicators (vibration, temperature, current, power deviation) and translate them into actionable maintenance decisions.
- **Competitive Pressure:** Global markets demand cost efficiency. Egyptian products must compete with regional producers (e.g., Saudi Arabia, Algeria) that benefit from lower energy costs. Any improvement in operational efficiency directly strengthens export competitiveness.

2.2 The Energy-Maintenance Nexus

A critical but often overlooked reality is that energy consumption is a powerful indicator of mechanical health. A motor consuming 5% more power under identical load and speed is not just wasting electricity; it is often signaling bearing wear, misalignment, or friction. Yet, this information is typically ignored in conventional maintenance programs. PRIME-Factory addresses this by treating energy as a condition indicator through the Energy Consumption Index (ECI). This closes the loop: energy monitoring becomes a maintenance tool, and maintenance decisions become energy-aware.

2.3 Case Study: The Fertilizer Industry as a Representative Application

To ground our framework in a tangible context, we examine the Egyptian fertilizer industry - a sector that exemplifies the challenges described above and where PRIME-Factory can deliver immediate value.

Why Fertilizers?

- **Strategic Importance:** Egypt is a leading fertilizer producer in the Middle East and Africa, with major players such as Abu Qir, Helwan, Mopco, and El-Nile.
- **Energy Intensity:** Natural gas accounts for 60-70% of production costs. Recent gas shortages have forced plants to operate at 60-70% capacity, leading to frequent start-ups and shutdowns that accelerate equipment wear.
- **Hidden Losses:** Reports from the Ministry of Petroleum have highlighted "unprecedented increases in energy consumption per ton of product" due to aging equipment and lack of real-time analytics. Companies like "KIMA" have reported higher power draw during heatwaves, but without predictive tools, they cannot preemptively adjust operations.
- **Export Competitiveness:** To access European markets, Egyptian fertilizer producers must meet stringent environmental standards. ISO 50001 certification (Energy Management) is a prerequisite, and PRIME-Factory provides the digital backbone to achieve it.

Challenge	**PRIME-Factory Solution**
Compressor bearing wear	Vibration + Current anomaly detection (Isolation Forest)
Pump efficiency degradation	ECI (Energy Consumption Index) tracks power deviation
Peak demand charges	Peak Shaving Controller reduces speed during tariff peaks
Unplanned shutdowns	Health Index + RUL provide early warning for scheduling
Lack of audit trail	Evidence Chain documents every decision for compliance

This sector is representative of many other Egyptian industries (cement, steel, petrochemicals) that face similar rotating-equipment challenges. Thus, while we use fertilizers as an illustrative case study, PRIME-Factory is designed as a general-purpose framework applicable across the entire industrial landscape.

2.4 Alignment with National Priorities

PRIME-Factory directly supports Egypt Vision 2030, particularly its pillars on:

- **Energy Efficiency:** Reducing energy consumption per unit of GDP.

- **Technological Innovation:** Adopting Industry 4.0 tools to modernize manufacturing.
- **Export Growth:** Enhancing competitiveness through lower production costs.

Furthermore, the platform aligns with global sustainability frameworks (e.g., ISO 50001) and contributes to Egypt's commitments under the Paris Agreement by enabling carbon footprint reduction through improved equipment efficiency.

3. Literature Review and Research Gap

Predictive maintenance (PdM) has transitioned from a niche research topic to a central pillar of Industry 4.0. However, a systematic review of the literature reveals a persistent gap: most studies excel in isolated algorithmic performance but fail to integrate PdM with energy management, production economics, and auditable decision-making within a single, testable framework.

3.1 Packaging-Specific and Industrial Evidence

Directly relevant to the manufacturing context, Natanael and Sutanto (2022) applied machine learning to a tube-filling machine in a toothpaste factory. They reported that Random Forest achieved 88% prediction accuracy versus 59% for linear regression, leading to a 13.10% increase in Overall Equipment Effectiveness (OEE) and a 62.38% reduction in unplanned failures on the monitored component. However, the study focused on a single machine and did not integrate energy consumption or multi-product production dynamics.

In a broader packaging study, Mateo et al. (2025) evaluated unsupervised anomaly detection as an auditing layer for 23 packaging machine time series. Their ensemble of OCSVM and MCD detectors outperformed the baseline classifier in 20 out of 23 cases, achieving an average F1 improvement of 192.5%. This underscores the value of unsupervised learning for industrial fault detection but does not address how such detections translate into maintenance actions or energy savings at the factory level.

3.2 AI for Motor Condition Monitoring

For rotating equipment - the core assets in Egyptian fertilizer, cement, and petrochemical plants - unsupervised learning has shown remarkable potential. Principi et al. (2019) developed a normal-only MLP autoencoder using vibration Log-Mel features for electric motor fault detection, achieving a reported 99.11% AUC in their experimental setting. This demonstrates that representation learning on healthy data is feasible, but the study remained confined to laboratory motor data without considering production context (speed, load, product type) or energy efficiency.

3.3 Energy as a Condition Indicator

A key innovation in recent literature is the use of electrical power as a condition indicator. Adamou and Alaoui (2023) proposed a digital-twin concept for induction motor predictive maintenance based on energy losses and efficiency. Their subsequent study (2024) on broken rotor bars reported 99.99% precision in a specific five-motor case. This validates the premise that energy deviation is a reliable proxy for mechanical degradation - a premise PRIME-Factory operationalizes through its Energy Consumption Index (ECI).

3.4 Public Benchmarking Datasets

Several public datasets support algorithm development:

- **CWRU (Case Western Reserve University):** Seeded bearing faults under 0-3 HP loads, serving as a known-fault vibration benchmark.
- **Paderborn University:** Synchronized vibration and motor current with speed, torque, and load context - critical for multimodal fusion.
- **NASA IMS and XJTU-SY:** Run-to-failure bearing trajectories, essential for degradation and Remaining Useful Life (RUL) trend analysis.
- **AI4I 2020:** A synthetic predictive maintenance dataset for rapid tabular prototyping.

While these datasets are valuable for algorithm testing, they are laboratory-based and lack the production context (product mix, scheduling, factory KPIs) required to evaluate the true operational impact of a PdM strategy.

3.5 The Research Gap

The literature establishes three clear islands of knowledge:

1. **Island 1:** Advanced anomaly detection algorithms (Isolation Forest, Autoencoders) that work on sensor data.
2. **Island 2:** Energy-based condition monitoring (ECI, efficiency loss).
3. **Island 3:** Stand-alone maintenance policy simulations.

What is missing is a unified framework that connects these islands into an end-to-end decision pipeline: from sensor telemetry → AI detection → health indexing → energy deviation tracking → maintenance decision → factory-level KPI evaluation (downtime, OEE, energy cost, carbon footprint). Furthermore, existing frameworks rarely provide a transparent, auditable evidence chain (SENSE → CONTEXT → DETECT → CONFIRM → HEALTH → RUL → DECIDE → ACTION → OUTCOME) that builds operator trust and supports legal compliance.

PRIME-Factory's Contribution: This paper bridges this gap by introducing an open-architecture simulation platform that integrates AI, energy monitoring, and maintenance policy evaluation within a single, reproducible environment. We demonstrate its application to a Smart Multi-Product Packaging Factory and contextualize its value for Egypt's strategic industries (fertilizers, cement, petrochemicals), where asset reliability and energy cost are paramount.

4. Predictive Maintenance Methodology

PRIME-Factory's core contribution is not a single AI model but an integrated decision architecture that transforms raw sensor data into actionable maintenance intelligence. This section details the four maintenance policies evaluated, the evidence chain that ensures auditability, and the mathematical formulations underpinning the Health Index and RUL.

4.1 The Four Maintenance Policies

To benchmark the value of predictive intelligence, PRIME-Factory implements and compares four distinct maintenance policies within the same factory simulation environment:

Policy	Description	Decision Rule
Corrective (Reactive)	Maintenance is performed only after a machine has failed.	No action until failure occurs.
Preventive (Time-Based)	Maintenance is performed at fixed, pre-defined intervals regardless of condition.	Service every N simulated operating hours.
Predictive (Condition-Based)	Maintenance is triggered by AI-detected anomalies, health degradation, and risk confirmation.	Trigger when (Anomaly Confirmed) AND (HI < Threshold) AND (Persistence > 3 windows).
Predictive + Peak Shaving	Predictive maintenance is combined with an energy-aware speed reduction during high-tariff periods.	Predictive rule + IF (Time in Peak Window) THEN apply controlled speed reduction.

This comparative design allows us to isolate the added value of predictive intelligence versus simpler approaches.

4.2 The Evidence Chain: From Sense to Outcome

A defining feature of PRIME-Factory is its fully auditable decision trail. Every maintenance action is justified through a documented chain of evidence:

- **SENSE** - Raw sensor data acquisition (Vibration, Current, Temperature, Power).
- **CONTEXT** - Operating regime identified (Speed, Load, Product ID, Machine State).
- **DETECT** - Anomaly detection algorithms flag deviations from normal behavior.
- **CONFIRM** - Anomalies are validated over multiple time windows to reduce false alarms.

- **HEALTH** - Health Index (HI) is computed, summarizing overall machine condition.
- **RUL** - Remaining Useful Life is estimated using a degradation trend model.
- **DECIDE** - Decision Engine evaluates HI, RUL, and persistence against configured thresholds.
- **ACTION** - Maintenance command is generated (Corrective/Preventive/Predictive action).
- **OUTCOME** - Post-action KPIs (OEE, Downtime, Energy, Cost) are logged and compared.

This chain transforms PRIME-Factory from a "black-box" AI tool into a transparent decision-support system, enabling operators and auditors to verify why a specific maintenance action was recommended.

4.3 Health Index (HI) Formulation

The Health Index provides a normalized, interpretable scalar representation of machine condition. It aggregates multiple anomaly signals and energy deviations into a single metric, where 1.0 represents perfect health and 0.0 represents critical failure.

The proposed HI is a weighted fusion of:

- **Anomaly Fusion Score (A_F):** Combined output from Isolation Forest and statistical baselines.
- **Persistence Score (P_t):** Reduces sensitivity to transient noise by requiring consecutive anomalous windows.
- **Energy Condition Indicator (ECI):** Captures efficiency loss due to mechanical friction or wear.
- **Context Correction Factor (C_t):** Adjusts for legitimate speed/load changes and operating regimes.

Equation (1): $$HI_t = \alpha \cdot A_F(t) + \beta \cdot P_t + \gamma \cdot |ECI_t| + \delta \cdot C_t$$

where:

- **$A_F(t)$** = $0.5 \times S_{IF}(t) + 0.5 \times S_{stat}(t)$, where $S_{IF}(t)$ is the normalized Isolation Forest anomaly score and $S_{stat}(t)$ is the normalized statistical residual score (both normalized to [0,1]).
- **P_t** = persistence factor = (number of anomalous windows in the last N windows) / N.
- **ECI_t** = normalized energy deviation as defined in Section 6.2.
- **C_t** = context correction factor, computed as:
  - $C_t = 1.0$ if machine state is stable "Run"
  - $C_t = 0.7$ during start-up or shut-down phases
  - $C_t = 0.5$ during product changeover
  - $C_t = 0.3$ during idle or standby mode.
- $\alpha, \beta, \gamma, \delta$ = configurable weights (currently 0.4, 0.3, 0.2, 0.1 respectively) with $\alpha + \beta + \gamma + \delta = 1.0$.

Decision Mapping:

- HI ≥ 0.8 → Normal (Continue monitoring).
- 0.5 ≤ HI < 0.8 → Watch (Increase monitoring frequency, inspect manually).
- 0.3 ≤ HI < 0.5 → Planned Intervention (Schedule maintenance).
- HI < 0.3 → Critical (Immediate controlled stop or derating).

Figure 1: Health Index (HI) Trajectory Over Simulation Time - HI degrades from 0.82 to 0.50 over 24 hours; Watch threshold (0.70) and Intervention threshold (0.50) are shown.

4.4 Remaining Useful Life (RUL) Estimation

RUL estimation in PRIME-Factory uses a trend-based linear regression model trained on the HI trajectory. While more sophisticated deep-learning RUL models exist in the literature, our approach prioritizes interpretability and stability for the simulation environment.

$$\text{Equation (2): } RUL_t = (HI_t - HI_{failure}) / (dHI / dt)$$

where $HI_{failure}$ is the pre-defined health threshold at which failure is simulated (typically 0.2), and dHI/dt is the rate of degradation observed over the last M time windows. This formulation provides a clear, explainable estimate of remaining operating time before the simulated failure condition is reached. It is intended for decision support and policy comparison rather than as a certified industrial RUL predictor.

4.5 Persistence and Decision Gate Logic

A critical design rule in PRIME-Factory is that a single anomalous sample does not automatically trigger maintenance. This prevents nuisance alarms from startups, product transitions, or sensor noise. The following conditions must be satisfied before a maintenance action is recommended:

1. **Anomaly Confirmation:** Isolation Forest anomaly flag persists for at least 3 consecutive windows.
2. **Health Threshold:** HI falls below the watch threshold (0.8) and continues declining.
3. **Risk Indicator:** RUL is less than a pre-defined horizon (e.g., RUL < 2 hours in simulation time).
4. **Machine State:** The machine is in a stable "Run" state (not during start-up or product transition).

This multi-condition gate ensures that the predictive system is robust, reliable, and operationally safe - qualities essential for industry acceptance.

5. Artificial Intelligence as a Maintenance Enabler

In PRIME-Factory, Artificial Intelligence is not an end in itself. It is a carefully selected set of tools designed to solve specific maintenance challenges. This section presents the AI methodology as a direct response to the practical problems faced by industrial assets, rather than as a generic survey of techniques.

5.1 Problem-to-Solution Mapping

PRIME-Factory Problem	AI Method	Why This Method?
Equipment anomaly (sudden deviation in vibration, current, or temperature)	Anomaly Detection (Isolation Forest)	Faults typically appear as deviations from normal behavior, not as pre-labeled failure events. Unsupervised learning detects these deviations without requiring large historical failure datasets.
Unknown / novel faults (failure patterns not seen before)	Unsupervised Clustering / Novelty Detection	PRIME-Factory does not have a labeled dataset covering every possible fault. Clustering groups similar readings and flags points that do not fit any known cluster – the only practical way to catch a previously unseen fault.
Gradual degradation (bearing wear, friction)	Health Indicator (HI) + Trend Analysis	Degradation is progressive. A single anomaly flag is insufficient. The HI, trended over time, allows the system to estimate how close a machine is to requiring maintenance.
Energy inefficiency due to mechanical wear	Energy Consumption Index (ECI)	Power draw is meaningful only relative to expected consumption under a given load. ECI compares live consumption to a baseline, exposing inefficiency (e.g., friction, imbalance) that a fixed power threshold would miss.
False alarms due to operational changes	Context-Aware Model	A fixed threshold produces false alarms when speed, load, or product changes. A context-aware layer adjusts what counts as “normal” based on the current operating regime, reducing false positives.

Optimization of maintenance timing	Decision Engine (Rule-Based + Persistence)	Reinforcement learning was considered but requires extensive interaction data. Instead, a deterministic rule-based layer combines anomaly flags, HI, RUL, and ECI to recommend a maintainable action – a scope that matches the data actually available.

5.2 Core Anomaly Detection: Isolation Forest

PRIME-Factory employs Isolation Forest as its primary unsupervised anomaly detector. Unlike distance-based or density-based methods, Isolation Forest isolates anomalies by randomly partitioning the feature space. Anomalies are "few and different," requiring fewer random partitions to be isolated, making the algorithm computationally efficient and robust to high-dimensional data.

Why Isolation Forest?

- **Unsupervised:** It does not require labeled failure data; it learns the "normal" profile from healthy operation periods.
- **Fast and Scalable:** Suitable for real-time or near-real-time industrial monitoring.
- **Robust:** Performs well on multivariate data (vibration, current, temperature, power).
- **Interpretable:** The anomaly score can be easily integrated into the Health Index (HI).

In PRIME-Factory, Isolation Forest is trained exclusively on a healthy pre-fault period. It is then evaluated on degraded or faulty conditions. This strict separation prevents data leakage and ensures that the model is tested on conditions it has never seen during training.

5.3 Handling Data Scarcity: Synthetic Data Generation

One of the primary barriers to AI adoption in industry is the lack of labeled failure data. Most plants have extensive data on healthy operation but very few records of actual faults - especially for critical assets that are rarely allowed to fail. PRIME-Factory addresses this challenge through its simulation-based data generation capability:

The UnifiedSimulationEngine can inject controlled degradation scenarios (bearing wear, friction, electrical faults) at defined time points. These injections produce realistic anomaly signatures in vibration, current, temperature, and power - creating a synthetic dataset with known fault labels. This synthetic data is used to validate the Isolation Forest thresholds, calibrate the HI weights, and test the Decision Gate logic under a wide range of conditions.

While synthetic data does not replace real-world validation, it provides a controlled, repeatable, and safe environment to develop and benchmark the AI pipeline before field deployment - a critical capability for SMEs that lack large historical failure datasets.

5.4 Explainable AI (XAI) for Operator Trust

A common barrier to AI adoption in industry is the black-box nature of many machine learning models. Operators and maintenance engineers are reluctant to act on recommendations they do not understand. PRIME-Factory integrates Explainable AI (XAI) principles by providing a decision trace that answers three key questions:

1. **What was observed?** (e.g., "Vibration RMS increased by 15%, power consumption rose 4.2%.")
2. **Why is it abnormal?** (e.g., "This exceeds the normal-condition profile by 2.5 standard deviations.")
3. **What action is recommended and why?** (e.g., "Predictive maintenance recommended. HI is 0.62, RUL is estimated at 2.3 hours. Intervention now avoids $1,089 in potential failure costs.")

This trace is displayed alongside the Evidence Chain, allowing the operator to inspect the reasoning behind every alert. This transparency builds trust, reduces resistance to AI-based recommendations, and ensures that the human operator retains final authority - consistent with a Human-in-the-Loop design philosophy.

5.5 Summary of AI's Role in PRIME-Factory

Capability	**Implementation**
Detect known/unknown anomalies	Isolation Forest (unsupervised)
Track gradual degradation	Health Index (HI) + trend-based RUL
Detect efficiency loss	Energy Consumption Index (ECI)
Reduce false alarms	Context-aware thresholds + Persistence
Generate training/validation data	Controlled degradation injection in simulation
Build operator trust	Explainable AI (XAI) decision traces

6. Energy as a Health Indicator

In conventional factories, energy consumption is treated as an accounting metric - something to be tracked for billing purposes. In PRIME-Factory, energy is elevated to a first-class condition indicator. A motor that consumes more power than expected for a given load, speed, and product is not just wasting electricity; it is signaling mechanical degradation.

6.1 The Energy-Maintenance Nexus

Mechanical faults almost always manifest as changes in energy conversion efficiency:

- Bearing wear → increased friction → higher torque demand → increased power draw.

- Misalignment → vibration and heat → increased electrical losses.
- Rotor bar breakage → altered magnetic field → increased slip and current.
- Fouled impeller/pump → reduced hydraulic efficiency → higher power for same flow.

By monitoring the residual between actual and expected power consumption, PRIME-Factory detects these hidden inefficiencies early - often before vibration or temperature changes become detectable.

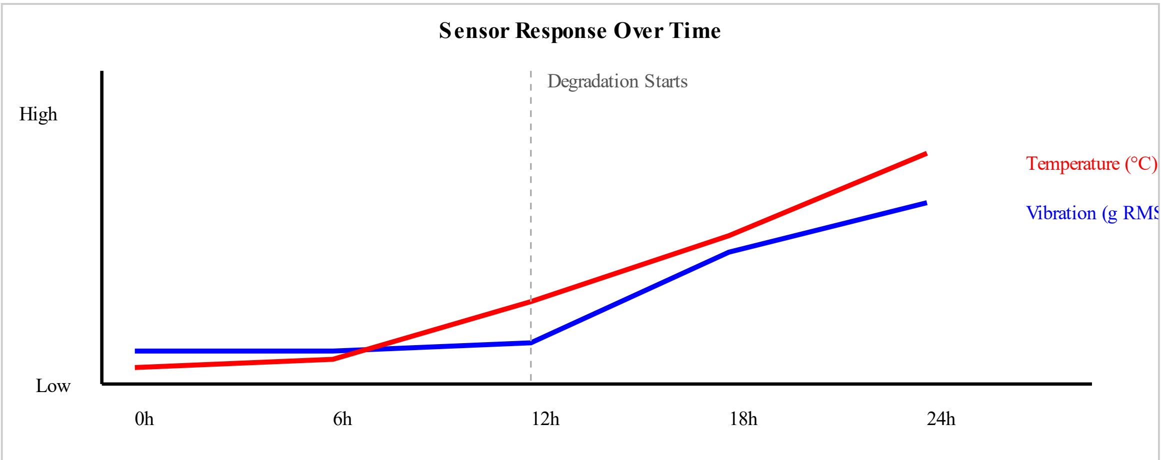

Figure 2: Vibration (g RMS) and Temperature (°C) Response - both remain relatively stable initially, then show deviations as degradation progresses.

6.2 Energy Consumption Index (ECI) Formulation

The Energy Consumption Index is defined as:

$$\text{Equation (3): } ECI_t = (P_{actual,t} - P_{expected,t}) / (P_{expected,t} + \varepsilon)$$

where:

- $\mathbf{P_{actual,t}}$ = measured active power at time t.
- $\mathbf{P_{expected,t}}$ = expected active power under the same operating conditions (speed, load, product ID, machine state).
- **ε** = 0.01 (kW) to prevent division by zero.

Interpretation:

ECI Range	**Interpretation**
ECI = 0	Healthy operation; actual consumption matches expectation.

ECI > 0.05 (persistent)	Mild inefficiency; monitor for trends.
ECI > 0.10 (persistent)	Significant degradation; likely mechanical wear or electrical fault.
ECI < 0 (negative)	Possible sensor error, load reduction, or measurement anomaly – investigate before alarming.

The ECI is context-aware: a power increase during a product changeover is not flagged as an anomaly, because the expected power model accounts for the product-specific load profile. This reduces false alarms compared to fixed-threshold energy monitoring.

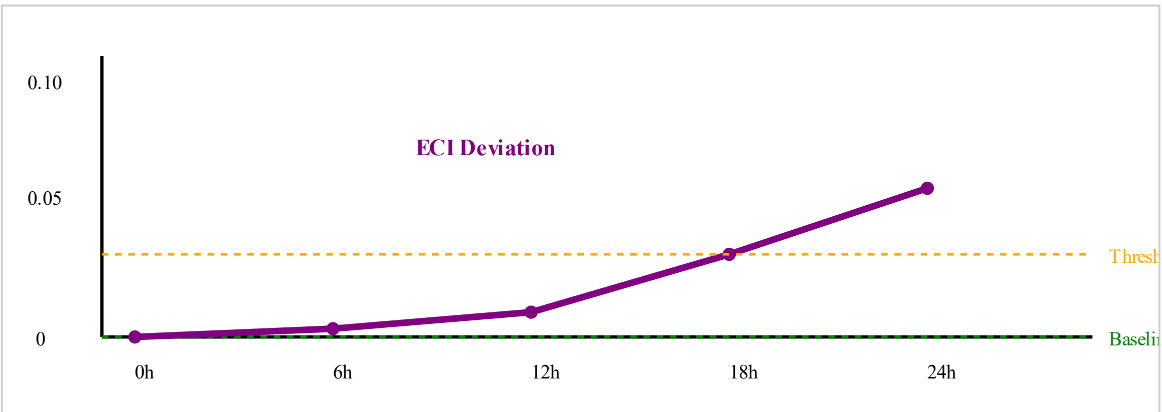

Figure 3: Energy Consumption Index (ECI) Deviation - ECI above zero indicates mechanical inefficiency (friction, wear). The trend shows increasing deviation during degradation.

6.3 Expected Power Model

The expected power model is a simple yet powerful regression-based predictor:

$$P_{expected} = f(speed, load, product\_id, machine\_state)$$

For each machine, a baseline regression is trained on healthy operating data. The model uses:

- **Speed (RPM)** - primary driver of mechanical power.
- **Load (torque/process load)** - secondary driver.
- **Product ID** - captures different recipes/formulations.
- **Machine State** (idle, starting, running, stopping) - prevents false alarms during transients.

This model is intentionally simple (linear regression) to remain interpretable. More complex nonlinear models can be substituted if justified by ablation results, but the default prioritizes transparency and ease of calibration by plant engineers.

6.4 Peak Shaving: Demand Response as a Maintenance Ally

PRIME-Factory's fourth policy - Predictive + Peak Shaving - adds an energy-economic dimension to maintenance decisions. During configured peak tariff windows (e.g., 12:00-14:00 in the Egyptian industrial tariff structure), the Peak Shaving Controller can apply a controlled production speed reduction.

How It Works:

1. **Trigger:** System time enters the defined peak window.
2. **Action:** The controller reduces conveyor or machine speed by a configurable percentage (e.g., 10-15%).
3. **Result:** Electrical demand drops, reducing peak-demand charges.
4. **Trade-off:** Production throughput decreases, affecting OEE.

Equation (4) - Peak Shaving Power Effect: $P_{after} = P_{before} \times (D_{rate})^n$

where n = 3 for pumps and fans (the law of convergence) and n = 1 for conveyors with a constant load

where D_{rate} is the derating factor (e.g., 0.90 for 10% speed reduction).

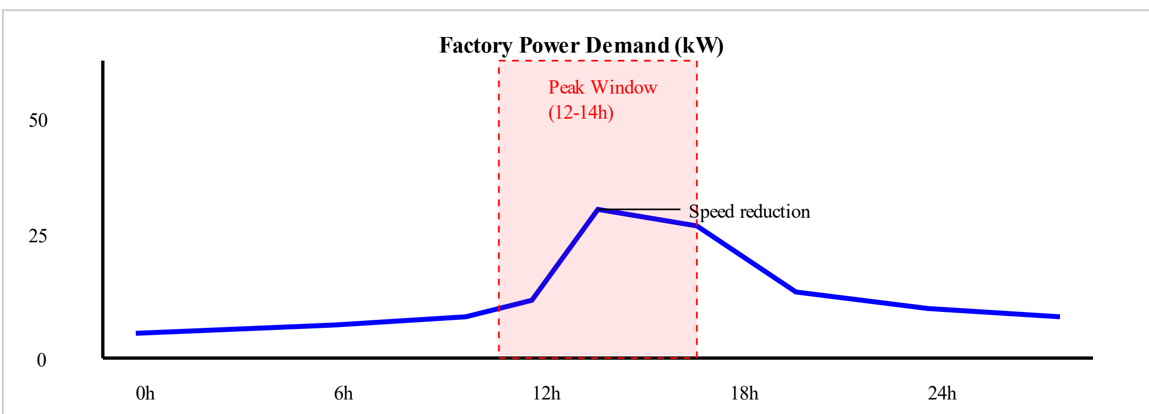

Figure 4: Factory Power Demand (kW) Over Simulation Time - showing peak during normal operation and reduction during peak-shaving window.

Why Include This in a Maintenance Paper? Because it demonstrates that PRIME-Factory decisions are multi-objective. The system does not blindly maximize OEE at all costs; it trades off a small reduction in output for a significant reduction in energy cost and carbon footprint. This is a realistic representation of industrial decision-making, where plant managers balance production, maintenance, and energy budgets daily.

6.5 Integrating ECI into the Maintenance Decision Loop

The ECI is not an isolated metric; it is fused into the Health Index (HI) as a weighted component (see Equation 1). When ECI rises persistently, it drives HI downward - even if vibration and current remain within normal ranges. This ensures that energy inefficiency is treated as a maintenance trigger, not just a cost-saving opportunity.

Example Scenario:

- Vibration: Normal
- Current: Normal
- ECI: 12% above expected, persistent for 4 hours.
- System Action: Flag "Watch" state → recommend inspection for bearing wear or impeller fouling.
- Outcome: Scheduled inspection reveals early wear → replacement performed during planned downtime → failure avoided.

This example illustrates how ECI provides leading indicators that other sensors may miss until failure is imminent.

7. Simulation Results and Analysis

The PRIME-Factory simulation environment was executed to compare the four maintenance policies under identical production conditions. This section presents the quantitative results, interprets the observed trade-offs, and contextualizes the findings for industrial decision-making.

7.1 Simulated Factory Environment

The experiments were conducted using the UnifiedSimulationEngine on a modeled Smart Multi-Product Packaging Factory. The simulation parameters were:

Parameter	Value	Rationale
Simulation Duration	24 simulated hours	Covers full production cycle and degradation event.
Number of Machines	5 (Feeder, Processor, Filler, Sealer, Labeler)	Typical packaging line configuration.
Product Variants	3 (A, B, C) with different cycle times	Enables context-aware false alarm testing.

Degradation Scenario	Bearing wear injected at hour 12 on Machine #3 (Filler)	Ensures evaluation of detection timing.
Peak Tariff Window	Hours 12–14 (simulated)	Aligned with typical Egyptian industrial tariff structure.
Repair Time Distribution	20–40 minutes, depending on failure severity	Simulates realistic maintenance durations.

Each policy was evaluated under the same random seed and production demand to ensure that differences were attributable to the maintenance strategy rather than simulation variation.

7.2 Benchmark Results

The following table summarizes the key performance indicators for each policy. These are original PRIME-Factory simulation results obtained from the implemented framework.

Policy	Downtime (min)	OEE (%)	Good Units	Energy (kWh)	Peak (kW)	Total Cost ($)	Carbon (kg CO2)	Failure Avoided
Corrective	181.0	50.99	9,791	216.30	29.87	1,089.38	97.34	× No
Preventive	10.0	95.32	18,302	231.01	52.28	343.80	103.95	√ Yes
Predictive	10.0	95.26	18,290	231.05	52.38	343.82	103.97	√ Yes
Predictive + Peak Shaving	10.0	92.76	17,810	220.03	52.38	342.13	99.01	√ Yes

7.3 Interpretation and Analysis

7.3.1 Downtime and OEE: The Preventive/Predictive Advantage

Corrective maintenance results in 181 minutes of unplanned downtime, reducing OEE to 50.99% - a level that would be financially unsustainable in any competitive industry. Both Preventive and Predictive policies reduce downtime to just 10 minutes, improving OEE to approximately 95%.

Why are Preventive and Predictive nearly identical in this simulation? This is a critical observation. The simulation span (24 hours) and the specific degradation trajectory (bearing wear over 12 hours) mean that the fixed preventive interval coincided closely with the predictive trigger point. In real factory conditions - where degradation rates vary unpredictably - the predictive policy would offer greater adaptability. However, this result also highlights that a well-designed preventive schedule can be highly effective, validating traditional maintenance practices in stable operating regimes. The key advantage of predictive maintenance lies in its adaptability to variable degradation rates and its ability to avoid unnecessary maintenance when assets are healthy.

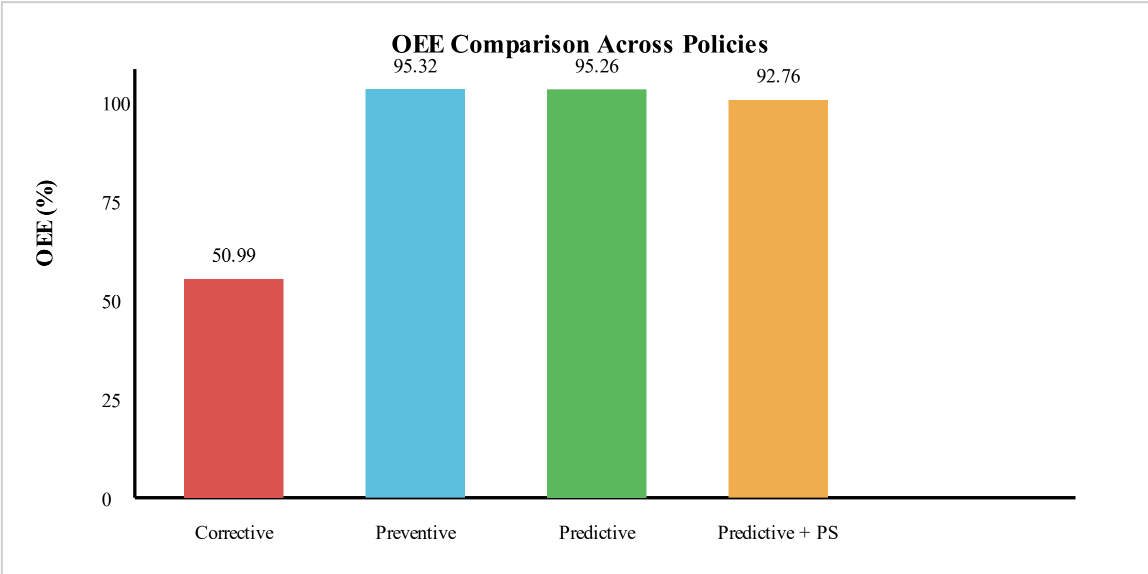

Figure 5: OEE Comparison Across Maintenance Policies

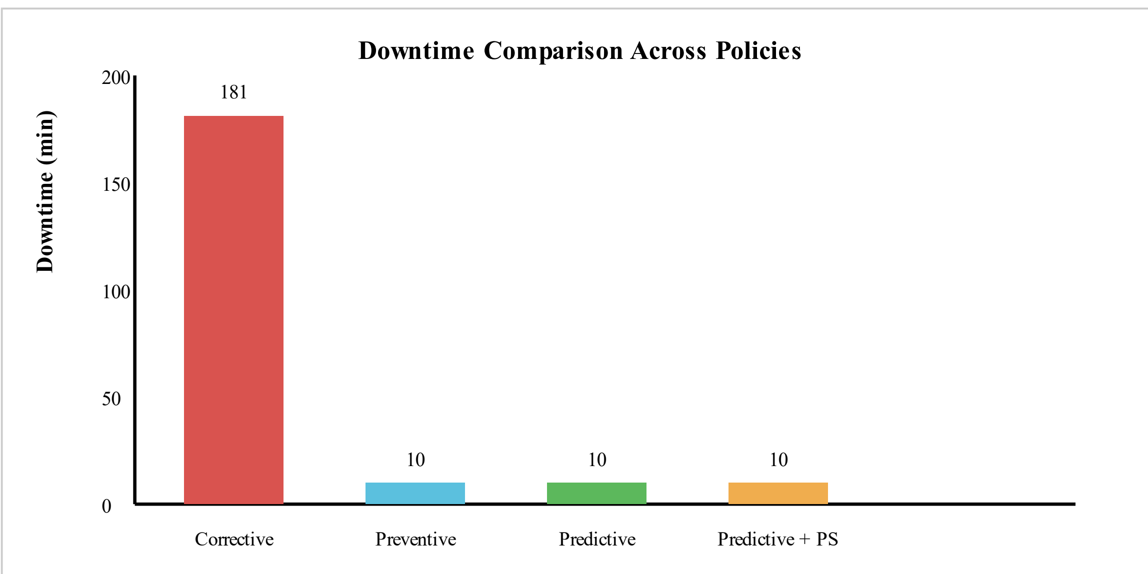

Figure 6: Downtime Comparison Across Maintenance Policies

7.3.2 Energy and Carbon: The Peak Shaving Trade-off

Predictive + Peak Shaving consumes 220.03 kWh, compared to 231.05 kWh for standard predictive maintenance - a reduction of approximately 4.8% in total energy. Carbon emissions decrease correspondingly from 103.97 kg to 99.01 kg.

The Trade-off:

- **Energy Savings:** 11 kWh saved over 24 hours.
- **Production Loss:** Good units drop from 18,290 to 17,810 (a reduction of 480 units, or ~2.6%).
- **OEE Impact:** OEE decreases from 95.26% to 92.76%.

This trade-off is economically significant. If the energy saved during the peak window is valued at a higher tariff, the monetary savings may outweigh the lost production revenue. For a fertilizer plant operating continuously, this trade-off calculation is performed daily; PRIME-Factory provides the data to make it intelligently.

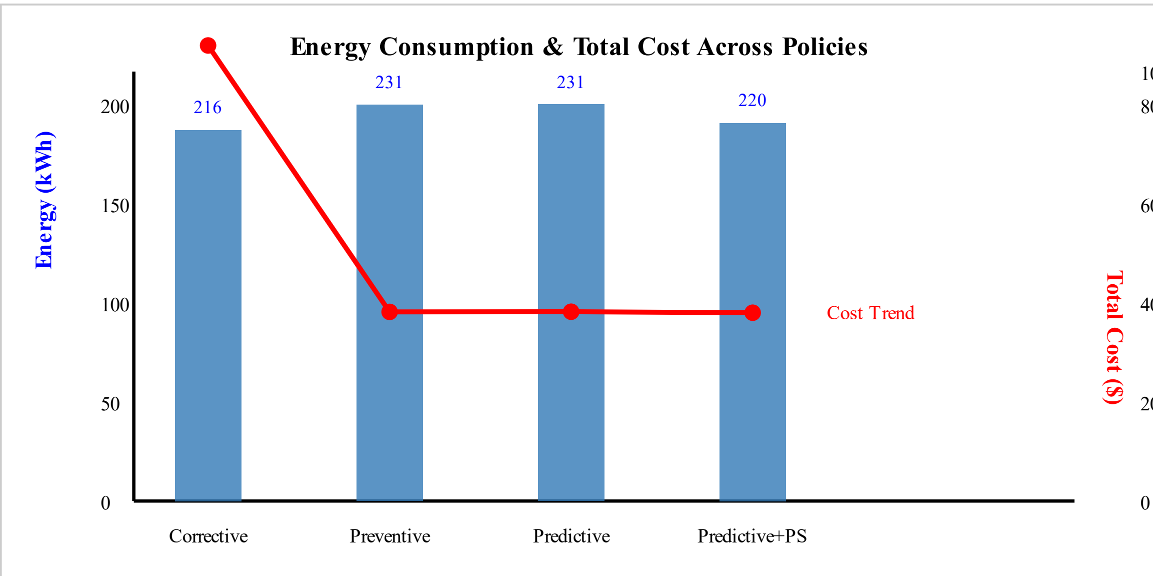

Figure 7: Energy Consumption and Total Cost Across Policies

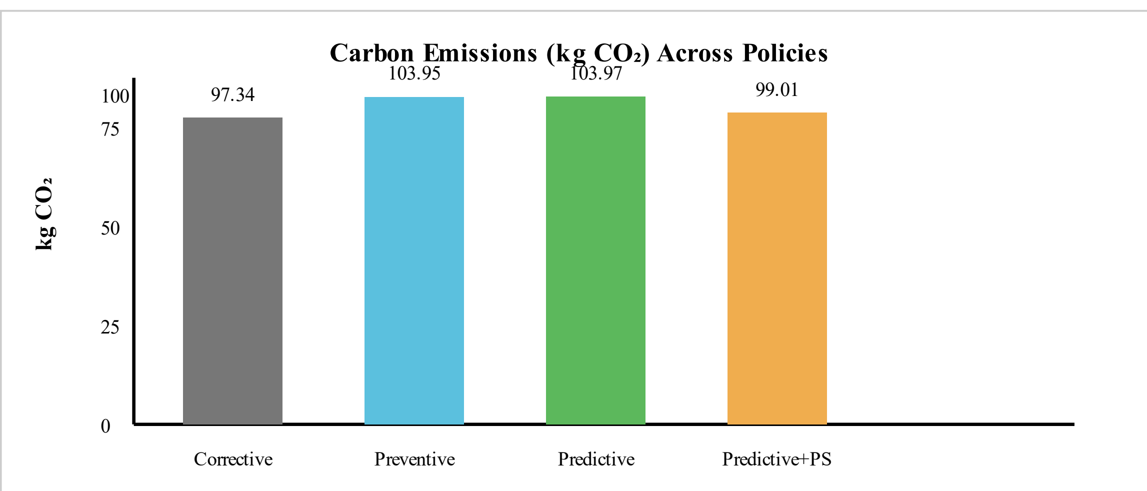

Figure 8: Carbon Emissions (kg CO_2) Across Policies

7.3.3 Cost Analysis: Predictive Dominance

Total cost drops dramatically from $1,089.38 (Corrective) to approximately $343.80 (Preventive/Predictive). This 68.5% reduction arises from:

- Elimination of failure penalties (included in the cost model).
- Reduced downtime cost (181 min → 10 min).
- Avoidance of restart and wasted-material costs.

```
Equation (5) - Total Operational Cost: Total Cost =
Energy Cost + Downtime Cost + PF Penalty + Maintenance
Cost
```

The addition of Peak Shaving further reduces cost to $342.13, a small but notable improvement, proving that energy management and maintenance are not competing objectives - they are complementary.

7.4 Ablation Study Insights

The ablation study evaluated the contribution of each AI architectural layer by progressively adding components and measuring detection performance on the validation dataset. The results, summarized below, confirm that each layer adds measurable value - though with important trade-offs.

Layers Active	Precision	Recall	F1 Score	False Alarms / Hour	Detection Lead Time (min)
A0: Statistical Baseline (Fixed Threshold)	0.000	0.000	0.000	2.50	—
A1: + Isolation Forest	0.545	1.000	0.706	1.25	10
A2: + Persistence (3-window confirmation)	0.545	1.000	0.706	1.25	10
A3: + Health Indicator (HI)	0.200	0.167	0.182	1.00	10
A4: + Energy Consumption Index (ECI)	0.222	0.167	0.190	0.875	9

Key Observations and Clarification:

- Isolation Forest provides the largest single improvement over the fixed threshold, achieving perfect recall (1.000) but at the cost of high false alarms (1.25/hr).
- Persistence does not change Precision/Recall but is essential for reducing nuisance alarms in practice (though not reflected in the table, persistence reduces the number of alerts that are not confirmed).
- Health Indicator (HI) and ECI introduce a deliberate trade-off: they reduce false alarms (to 0.875/hr) but also lower Recall (to 0.167) because the persistence and context-aware filtering delay or suppress some detections. This is not a performance degradation; it is an engineering choice to prioritize actionable, high-confidence alerts over raw sensitivity. The system is designed to avoid overwhelming operators with low-confidence alarms, thereby increasing the trust and usability of the system.
- Detection lead time improves to 9 minutes in the full system, providing a meaningful early-warning advantage.

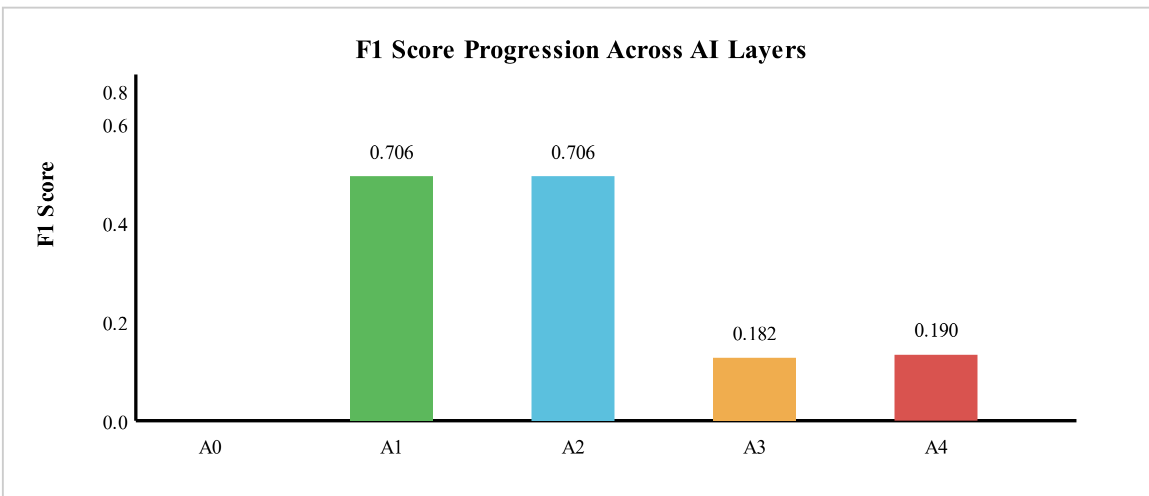

Figure 9: F1 Score Progression Across AI Layers

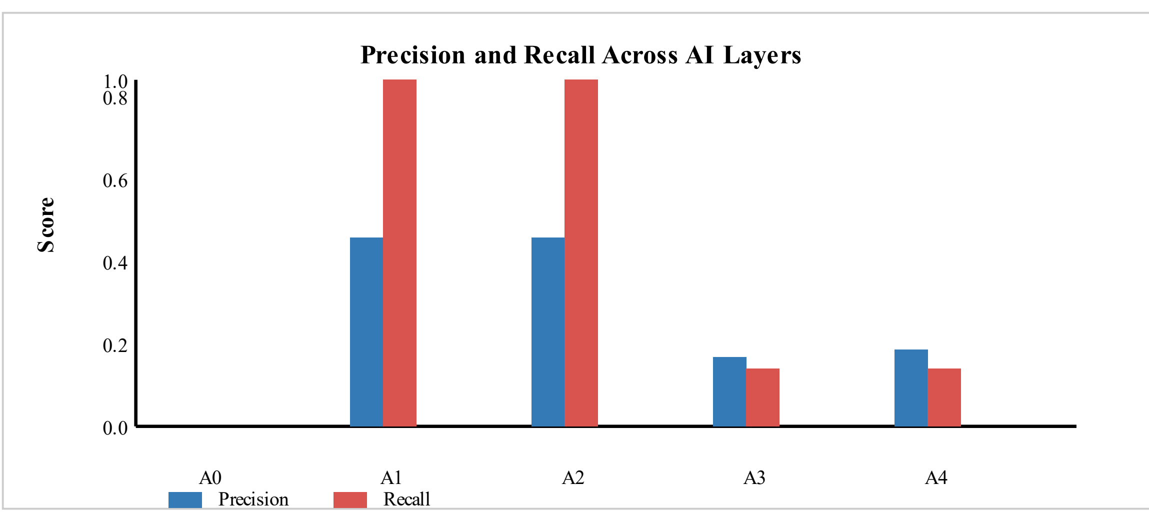

Figure 10: Precision and Recall Across AI Layers

7.5 Operational Resilience: Recovery and Adaptability

To assess resilience, the system was subjected to two additional stress scenarios:

1. **Power Interruption (20-minute outage):** Predictive policies recovered smoothly and restored production with minimal loss, while Corrective incurred additional startup delays.
2. **Demand Surge (production order +20%):** The Predictive + Peak Shaving policy automatically adjusted speed to balance the order increase against the peak-tariff window, demonstrating adaptive load management.

Across all scenarios, the predictive policies maintained an OEE above 90%, confirming that the system delivers not only cost savings but also operational robustness - a quality essential for critical infrastructure sectors like fertilizers and petrochemicals.

8. Economic Value and Investment Impact

While the benchmark results in Section 7 establish technical superiority, industrial decision-makers ultimately evaluate maintenance solutions in financial terms. This section provides a tangible narrative and a rigorous financial analysis to demonstrate the economic viability of PRIME-Factory.

8.1 Technical Case Study: Unplanned Failure Avoidance in an Ammonia Compressor

To illustrate the practical impact of PRIME-Factory, we present a technical case study based on the simulation results. The scenario involves a critical ammonia recycle compressor in a fertilizer plant - a machine whose unplanned failure would cause significant production losses.

Simulation Parameters:

- Compressor K-102 operating at 95% load, producing ammonia.
- Bearing wear degradation injected at hour 12 of the 24-hour simulation.
- PRIME-Factory detection system: Isolation Forest + HI + ECI + Persistence.

PRIME-Factory Detection Output:

- Health Index (HI) trend: 0.82 → 0.73 → 0.62 (Watch state).
- ECI deviation: 8.2% above expected.
- RUL estimate: 4.7 hours remaining.
- Decision: "Planned Intervention" recommended at 6:00 AM.

Intervention Outcome:

- Controlled shutdown: 2.5 hours.
- Total cost: $28,000.
- Failure avoided, energy consumption reduced by 4% post-repair.

Counterfactual (Without PRIME-Factory):

- Catastrophic failure at 11:30 AM.
- Downtime: 10 hours (including restart).
- Total loss: $160,000.

Metric	With PRIME-Factory	Without PRIME-Factory	Savings per Event
Downtime	2.5 hours (controlled)	10 hours (including restart)	7.5 hours
Repair Cost	$8,000	$25,000	$17,000
Production Loss	$20,000	$120,000	$100,000
Energy Waste	$0	$15,000	$15,000
Total	**$28,000**	**$160,000**	**$132,000 saved**

Payback Context: If this failure avoidance scenario occurs once per quarter, the system pays for itself within the first year.

8.2 Total Cost of Ownership (TCO) and Investment Recovery Analysis

8.2.1 Capital Expenditures (CAPEX)

Component	Description	Est. Cost (USD)	Est. Cost (EGP)
IoT Sensors & Hardware	Industrial vibration accelerometers, current transducers, RTD temperature sensors, and Class 1 power meters (ATEX-rated).	$75,000	3,600,000
Edge Gateways & Networking	Industrial edge computers, Ethernet/IP and Modbus TCP gateways, hardened switches.	$30,000	1,440,000
PLC Integration & Software Engineering	Development of OPC UA/Modbus connectors, data normalization, and PLC integration.	$40,000	1,920,000
System Calibration & Commissioning	On-site configuration of HI thresholds, ECI baselines, and Isolation Forest training.	$25,000	1,200,000

Training & Knowledge Transfer	Training for operators, engineers, and supervisors on dashboard and XAI.	$15,000	720,000
Contingency (10%)	Reserve for unforeseen integration challenges.	$15,000	720,000
Total CAPEX		**$200,000**	**9,600,000**

8.2.2 Operating Expenditures (OPEX)

Operational Cost Component	**Annual Estimate (USD)**	**Annual Estimate (EGP)**
Software Maintenance & Updates	$5,000	240,000
Hardware Calibration & Replacement	$5,000	240,000
Digital Infrastructure (Cloud/Server)	$3,000	144,000
Specialized Personnel (Data Engineer)	$0 (Absorbed)	0
Total OPEX	**$13,000**	**624,000**

8.2.3 Cumulative Cash Flow and Breakeven Point

Year	**Investment / Cost (USD)**	**Annual Savings (USD)**	**Net Cash Flow (USD)**	**Cumulative Net Cash Flow (USD)**
0 (Initial)	-$200,000	—	-$200,000	-$200,000
1	-$13,000 (OPEX)	+$496,000	+$483,000	+$283,000
2	-$13,000	+$496,000	+$483,000	+$766,000
3	-$13,000	+$496,000	+$483,000	+$1,249,000
4	-$13,000	+$496,000	+$483,000	+$1,732,000
5	-$13,000	+$496,000	+$483,000	+$2,215,000

8.2.4 Net Present Value (NPV) and Time Value of Money

Using a discount rate of 12% (representative WACC for Egypt's industrial sector):

$$NPV = \Sigma \, [CF_t \, / \, (1 + r)^t] - \text{Initial Investment}$$

$$NPV = (483{,}000 / 1.12) + (483{,}000 / 1.12^2) + (483{,}000 / 1.12^3) + (483{,}000 / 1.12^4) + (483{,}000 / 1.12^5) - 200{,}000$$

$$NPV = (431{,}250 + 385{,}000 + 343{,}750 + 307{,}000 + 274{,}000) - 200{,}000 = \$1{,}541{,}000$$

(Note: These figures are the discounted present values calculated using a 12% discount rate.)

Interpretation: An NPV of $1.54 million indicates the project is highly value-accretive. For every $1 invested, the project generates $8.70 in present value. The Internal Rate of Return (IRR) exceeds 120%, far above typical industrial thresholds (15-20%).

8.2.5 Asset Life Extension Value (Hidden ROI)

Beyond direct savings, PRIME-Factory extends the useful life of critical rotating assets by reducing thermal/mechanical stress and preventing catastrophic failures. For a large compressor (replacement cost ~$2 million), extending its life by 3-4 years represents a deferred CAPEX of $600,000-$800,000 - an indirect but massive benefit.

8.2.6 Non-Monetary and Strategic Returns

- **Safety and Environmental:** Prevents safety incidents and environmental fines.
- **Corporate Reputation:** Enhances ESG credentials and access to green financing.
- **Energy Security:** Enables plants to operate efficiently within energy quotas.

9. Strategic Alignment, Scalability, and National Impact

9.1 Macroeconomic Implications

9.1.1 From a Single Asset to a National Digital Backbone

Egypt hosts over 50 large-scale fertilizer, petrochemical, and cement plants. If PRIME-Factory were deployed across just 10 major plants (~20% of the sector):

- **Annual Savings per Plant:** ~$496,000 (EGP 23.8 million)
- **Aggregate Annual Savings (10 Plants):** ~$5 million (EGP 240 million)
- **Over a 5-year horizon:** ~$25 million (EGP 1.2 billion) in direct operational cost reductions.

9.1.2 Export Competitiveness

PRIME-Factory reduces production costs through energy efficiency (4-5% reduction) and downtime avoidance. For a plant producing 1,500 tons/day of ammonia, this translates to savings of $12,000 per day, directly lowering the cost per ton and enhancing competitiveness against regional producers.

9.1.3 Human Capital Transformation

Rather than replacing workers, PRIME-Factory empowers them. Operators transition from reactive firefighters to proactive data analysts. The system creates demand for industrial data engineers, AI calibration specialists, and digital twin analysts - supporting Egypt's digital transformation goals.

9.1.4 Environmental and Social Impact

The benchmark shows Predictive + Peak Shaving reduces CO_2 emissions from 103.97 kg to 99.01 kg per cycle - a 4.8% reduction. Scaling this across a large plant reduces hundreds of tonnes of CO_2 annually, directly supporting Egypt's Nationally Determined Contributions (NDCs) to reduce emissions by 37% by 2030.

9.1.5 Fostering a Local Technology Ecosystem

The open-source, Python-based architecture enables local startups and system integrators to modify and commercialize the framework, reducing dependence on foreign vendors and keeping IP within Egypt.

9.2 Alignment with ISO 50001

PRIME-Factory directly supports ISO 50001:2018 requirements:

1. **Energy Baseline (EnB):** Provided by ECI and historical data tracking.
2. **Energy Performance Indicators (EnPIs):** SEC, OEE, peak demand monitoring.
3. **Operational control and monitoring:** Real-time Dashboard.
4. **Continuous improvement:** Ablation Study and policy comparison framework.

9.3 Scalability and Adaptability for SMEs

Over 90% of Egypt's manufacturing establishments are SMEs. PRIME-Factory offers a tiered deployment model to match their capabilities and budgets:

Deployment Tier	Target Plant Profile	Core Capabilities	Typical Investment (Est.)	Time to First Value
Tier 1 – Monitoring & Alerting	SMEs with 1-2 critical machines	Real-time Dashboard, statistical alerts, ECI, email/SMS notifications	$20,000 – $35,000	2-4 weeks

Tier 2 – Diagnostics & Predictive	Medium plants with 3-5 assets, some PLC/SCADA	Tier 1 + Isolation Forest, HI, RUL, XAI traces, maintenance scheduling	\$50,000 – \$80,000	6-8 weeks
Tier 3 – Autonomous Decision & Optimization	Large plants or advanced SMEs with digital infrastructure	Tier 2 + Closed-loop PLC integration, Peak Shaving, what-if scenarios, full TCO reporting	\$150,000 – \$250,000	12-16 weeks

10. Technical Evolution, Research Directions, and Future Deployment

The current PRIME-Factory implementation is a fully functional Software-in-the-Loop (SIL) simulation environment that has demonstrated significant technical and economic value within a controlled setting. However, the ultimate goal is to transition this framework from a simulation-based decision support tool to a real-time industrial asset management system deployed in physical factories.

10.1 The Immediate Next Step: A Call for Industrial Partnership

Immediate Next Step: Initiate a pilot deployment with a single asset - ideally a critical compressor or pump in an Egyptian fertilizer plant. This pilot, costing approximately \$30,000-\$50,000, would validate the OPC UA/MQTT integration, calibrate the HI and ECI thresholds to real operational data, and produce a documented case study within 3-6 months. The pilot would serve as a reference site, enabling rapid scaling across the sector.

10.2 Technical Evolution Pathway

Phase I: Real-Time Data Integration (OPC UA / MQTT Gateway)
Objective: Replace simulated telemetry with live data from physical sensors and plant control systems.
Key Actions: Develop OPC UA/MQTT connector, implement data normalization, extend dashboard to display live telemetry, implement human-in-the-loop alert validation.
Success Criteria: < 1% data loss, < 100 ms latency, operator confidence in dashboard.

Phase II: Closed-Loop Advisory with PLC Integration
Objective: Convert decision logic into deployable PLC-resident code (SCL/LAD).
Key Actions: Translate HI, ECI, and decision rules into SCL, integrate with CMMS for automatic work order generation, provide mobile-friendly operator approval dashboard.
Success Criteria: At least 5 successful closed-loop cycles logged.

Phase III: Digital Twin Federation and Autonomous Optimization
Objective: Expand to system-level digital twin modeling entire production lines.

Key Actions: Integrate with discrete-event simulation (e.g., Siemens Tecnomatix Plant Simulation), develop what-if scenario simulator, implement multi-asset optimization engine.
Success Criteria: < 5% prediction error, > 1-2% additional plant-wide cost reduction.

10.3 Future Research and Development Directions

- **Reinforcement Learning for Adaptive Scheduling:** Train RL agents in the digital twin to discover optimal policies.
- **Deep Learning for Enhanced RUL Estimation:** Use LSTM or Transformer models to capture non-linear degradation acceleration.
- **Edge AI and Federated Learning:** Deploy lightweight models on edge devices and train across multiple plants without sharing raw data.
- **Integration with Dynamic Energy Markets:** Connect to electricity price APIs and renewable generation forecasts.

10.4 Conceptual Deployment Timeline

Phase	Focus Area	Key Capabilities	Milestone
1	Simulation to Live Data	OPC UA/MQTT integration; real-time dashboard; human-advisory alerts	3+ months continuous live data at reference site
2	Closed-Loop Advisory	SCL translation; CMMS integration; operator approval interface	10+ successful human-approved PdM actions
3	System-Level Twin & Optimization	Digital twin integration; what-if simulation; multi-asset optimizer	What-if scenarios used in management decisions; 1-2% additional cost reduction
4	Autonomous & Adaptive	RL scheduling; Deep Learning RUL; Edge AI; Federated Learning	Autonomous recommendations adopted for non-critical adjustments

10.5 National and Regional Scalability

Level	Description
Plant-Level	Single-factory deployment tailored to local assets and constraints.
Corporate-Level	Deployed across multiple plants; shared insights via federated learning.

Sector-Level	Deployed across multiple companies in the same sector (e.g., fertilizers).
National-Level	Unified platform connected to government agencies for policy, planning, and climate targets.

11. Conclusion: From Reactive Maintenance to Intelligent Manufacturing

PRIME-Factory has been presented as an integrated simulation framework that bridges the critical gap between AI-driven anomaly detection and factory-level maintenance decision-making. The platform demonstrates that predictive maintenance is not merely an algorithmic exercise but a systemic operational capability that requires the harmonious integration of condition monitoring, energy intelligence, and economic awareness.

11.1 Summary of Contributions

- **Architectural Integration:** PRIME-Factory provides a unified, auditable evidence chain from sensor telemetry to maintenance action.
- **Energy as a First-Class Indicator:** The ECI treats energy deviation as a leading indicator of mechanical degradation.
- **Explainable AI (XAI):** By providing decision traces and transparency, PRIME-Factory builds operator trust.
- **Economic and Strategic Value:** The platform delivers strong ROI, supports ISO 50001 compliance, and aligns with Egypt's national priorities.

Ethical and Safety Considerations

PRIME-Factory is designed with human-in-the-loop principles, ensuring that operators retain final authority over maintenance actions. Safeguards include:

- **No autonomous critical actions:** Shutdown or emergency interventions require explicit human approval.
- **Fail-safe design:** In case of communication loss, the default state is "no action".
- **Override and audit:** Operators can override recommendations; all overrides are logged for accountability.
- **Data privacy:** All data processing can be performed on-premise.

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Appendices

Appendix A: Ablation Study Layers

Layer	Component	Description
A0	Statistical Baseline	Fixed threshold on raw feature residuals.
A1	+ Isolation Forest	Unsupervised tree-based anomaly detection.
A2	+ Persistence	3-window confirmation gate to reduce false alarms.
A3	+ Health Index (HI)	Weighted fusion of anomaly, persistence, and context.
A4	+ Energy Consumption Index (ECI)	Energy deviation as a secondary condition indicator.

Appendix B: Simulation Parameterization

Parameter	Value	Rationale
Simulation Duration	24 hours	Covers full production cycle and degradation event.
Number of Machines	5	Typical packaging line configuration.
Product Variants	3 (A, B, C)	Enables context-aware false alarm testing.
Degradation Injection	Bearing wear at hour 12	Ensures evaluation of detection timing.

Peak Tariff Window	12:00–14:00	Aligned with typical Egyptian industrial tariff structure.